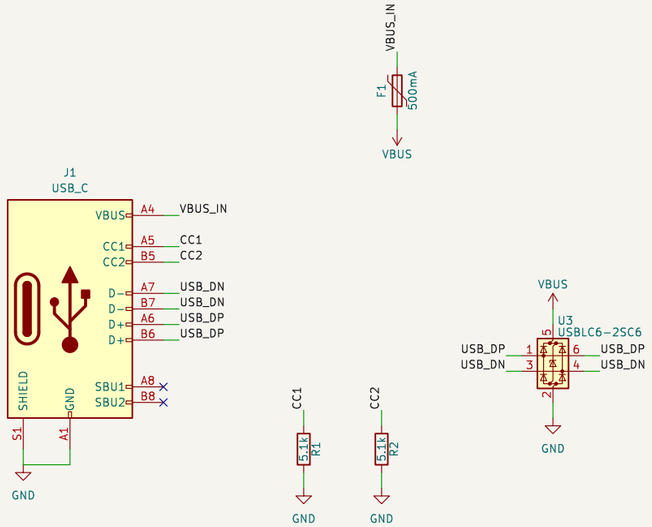
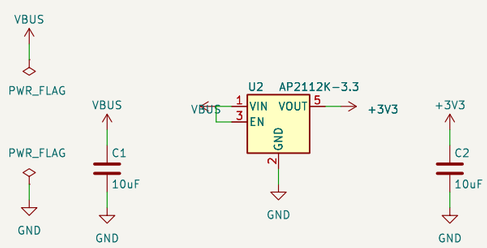
USB-C INPUT & PROTECTION



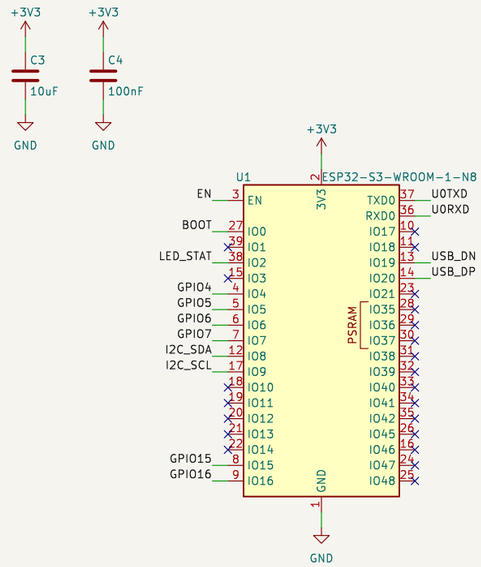
CC1/CC2 5.1k pull-downs = USB-C sink, up to 5V/3A advertised.
USBLC6 clamps ESD on D+/D- right at the connector.

3.3 V LDO REGULATOR



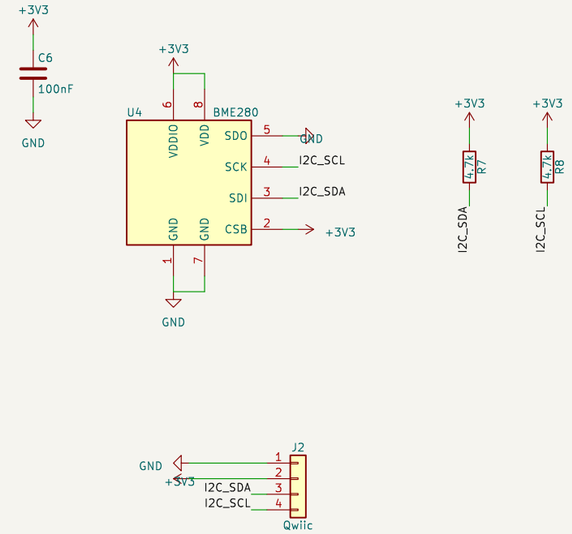
AP2112K: 600mA LDO, 250mV dropout. ESP32-S3 Wi-Fi TX peaks ~355mA.

ESP32-S3 MCU (native USB)



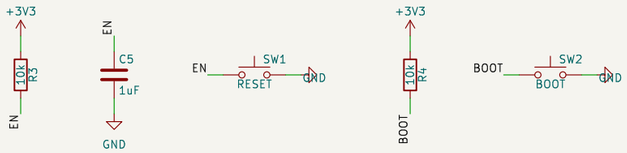
GPIO19/20 = native USB D-/D+ (flashing + CDC serial, no USB-UART chip).
IO0 low at reset = download mode.

BME280 SENSOR & QWIIC I2C

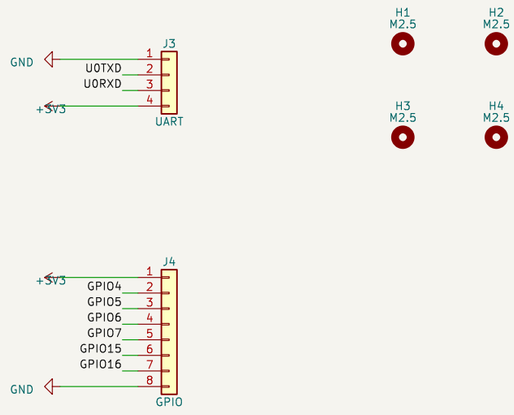


BME280 in I2C mode (CSB high), SDO low -> address 0x76. Qwiic pinout: GND, 3V3, SDA, SCL.

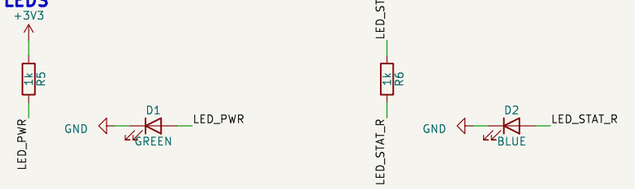
RESET / BOOT



DEBUG & GPIO EXPANSION



STATUS LEDS



Drawn in KiCad 7
USB-C powered, native USB, BME280 + Qwiic I2C, 2-layer 1.6mm
Muhammad Habban

Sheet: /
File: airnode.kicad_sch

Title: AirNode S3 – ESP32-S3 Environmental Sensor Node

Size: A3	Date: 2026-11-07	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1